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## Assignment # 2: Face Recognition

### 1 Problem Statement

We intend to perform face recognition. Face recognition means that for a given image you can tell the subject id. Our database of subject is very simple. It has 40 subjects. Below we will show the needed steps to achieve the goal of the assignment.

### 2 Download the Dataset and Understand the Format

1. ORL dataset is available at the following link. <https://www.kaggle.com/kasikrit/att-database-of-faces/>
2. The dataset has 10 images per 40 subjects. Every image is a grayscale image of size 92x112.

### 3 Generate the Data Matrix and the Label vector

1. Convert every image into a vector of 10304 values corresponding to the image size.
2. Stack the 400 vectors into a single Data Matrix  $D$  and generate the label vector  $y$ .  
The labels are integers from 1:40 corresponding to the subject id.

### 4 Split the Dataset into Training and Test sets

1. From the Data Matrix  $D_{400 \times 10304}$  keep the odd rows for training and the even rows for testing. This will give you 5 instances per person for training and 5 instances per person for testing.
2. Split the labels vector accordingly.

### 5 PCA Implementation (Your implementation)

1. Retain a percentage of variance using different thresholds:  $\alpha = \{0.8, 0.85, 0.9, 0.95\}$ .
  2. Project the dataset into the reduced PCA subspace.
  3. Report the dimensionality and visualize a few sample faces in PCA space.
- Hint. You can compute the eigenvalues once and save it in .npy file and load it afterwards.



## 6 Unsupervised Clustering

### a. K-Means Clustering (Your implementation)

1. Apply K-Means on the reduced PCA data for each alpha value.
2. Set number of clusters  $K = 20, 40, 60$ .
3. Evaluate using clustering accuracy.  
Plot (or tabulate) the performance measure (accuracy) against the K value and another one for the alpha value
4. Can you find a relation between alpha and classification accuracy?
5. Can you find a relation between K and classification accuracy?

### b. Gaussian Mixture Model Clustering (Your implementation)

1. Implement or use GMM clustering on the PCA outputs.
2. Again, use  $K = 20, 40, 60$ .  
Plot (or tabulate) the performance measure (accuracy) against the K value and another one for the alpha value
3. Can you find a relation between alpha and classification accuracy?
4. Can you find a relation between K and classification accuracy?

## 7 Evaluation

1. Evaluate the best K-Means model on the test set using the resultant centroids. Evaluate using accuracy, F1-Score and confusion matrix.
2. do the same thing for the best GMM.
3. Compare the performance of GMM to K-Means.

## 8 Bonus

- Implement a simple autoencoder for dimensionality reduction and apply K-Means and Gaussian Mixture Model (GMM) on the reduced space. Compare against other results obtained.
- Show samples of reconstructed images from the autoencoder



## 9 Submission Notes

- Work in groups of 3 students.
- Submit a **Jupyter Notebook** or a well-commented Python script containing your code.
- Submit a **PDF report** detailing your approach, results, and analysis.
- Ensure reproducibility by setting random seeds and structuring code properly.

### Note

- While discussions are allowed, all code and reports must be your own work. Plagiarism will result in a penalty.